

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I Gangisetty Nithin kumar(192325048) certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G nithin kumar



CSA0716 – Computer Networks

MACHINE LEARNING FOR NETWORK RELIABILITY

Machine Learning-Based Error Detection in Networks

A technical presentation on applying supervised learning models to detect, classify, and predict network anomalies in real time

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Under the Guidance :
D Aberna



INTRODUCTION

The Problem with Modern Networks

Today's networks carry massive, fast-changing traffic across distributed devices. Faults, such as congestion, packet loss, hardware failure and intrusion, surface as subtle patterns hidden in traffic data long before they cause visible outages.



Rule-based systems are static

Fixed thresholds can't adapt to evolving traffic behavior or novel fault types.



Manual monitoring doesn't scale

Human-driven diagnosis is too slow for networks generating millions of events per second.



Faults are often silent

Many errors degrade performance gradually instead of triggering an alarm.



Why It Matters

43%

of network downtime traces back to faults detected too late

<50ms

target detection latency for real-time ML pipelines

1000x

growth in monitored traffic volume over the last decade

MOTIVATION

Why Machine Learning Fits This Problem



Learns from data

Detects complex, non-linear traffic patterns that fixed rules and thresholds cannot express.



Adapts over time

Retrains on new traffic to keep pace with evolving usage patterns and emerging fault types.



Predicts, not just reacts

Flags early warning signals before faults escalate into full outages.



Improves accuracy

Reduces false positives compared to static threshold-based alerting systems.

COMPARISON

Traditional Detection vs. ML-Based Detection

Traditional (Rule-Based)

- Fixed thresholds set manually by engineers
- Cannot generalize to unseen fault patterns
- High false-positive / false-negative rate
- Requires constant manual tuning
- Struggles with high-dimensional, high-volume data

Machine Learning-Based

- ✓ Learns thresholds and patterns directly from traffic data
- ✓ Generalizes to novel and evolving fault types
- ✓ Lower false-positive rate via probabilistic scoring
- ✓ Retrains automatically as new data arrives
- ✓ Scales to high-dimensional, streaming network data



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